

Energy-Efficient UAV Trajectory Generation Based on System-Level Modeling of Multi-Physical Dynamics

Nicolas Michel¹, Zhaodan Kong¹, Xinfan Lin¹

Abstract—Electric multirotor aircraft is a major emerging technology, but currently limited by energy performance, which affects flight time and range. In this paper, energy-efficient trajectory generation is performed based on a system-level model of the multi-physical dynamics of an octorotor vehicle. An energy-optimal control problem is formulated using the model, and solutions are calculated and discussed for a range of hover-to-hover flight operations, which include both vertical and horizontal components. To allow for rapid, flexible, and computationally efficient generation of target state trajectories for real-time control applications, polynomial equations are used to approximate the optimized trajectories, which are segmented and normalized based on features of the optimized behaviors. The energy performance of the polynomial trajectories is compared to that of the original optimal trajectories in simulation with feedback control, demonstrating very close energy consumption. Additionally, a baseline PID controller is defined for comparison, and is shown to result in an increased energy use of 11.2% over feedback control using the optimized trajectories. The performances of each controller for a sample operation is compared, and behaviors which affect energy performance are discussed to demonstrate the value of the model-based trajectory generation approach.

I. INTRODUCTION

Multirotor unmanned aerial vehicles (UAVs) are a technology with significant potential for a wide range of applications, including package delivery, military Intelligence, Surveillance, and Reconnaissance (ISR) missions, and even passenger-scale Urban Air Mobility (UAM). However, energy performance is a critical challenge for each of these applications, which generally limits flight time to around 30 minutes for small (centimeter- to decimeter-scale) multirotor UAVs [1]. This is insufficient for current UAV practice, and presents a significant challenge for future applications. For example, it is estimated that UAM will require flight times up to 1 hour with a range of over 70 miles, with especially high power requirements during the vertical take-off and landing periods [2]. Meeting such performance specifications would require energy systems with an energy density of 350-400 Wh/kg, while the currently predominant lithium ion battery technology can only provide around 150 Wh/kg at the system level [2]. In addition to improvements in battery technology, bridging this gap will require research on improving the energy performance of multirotor.

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One widely studied topic is to optimize the multirotor operational energy efficiency through model-based planning and control [3]–[6]. Several papers in literature have contributed to this subject, but there is a lack of work which uses a complete system-level multirotor UAV model to exploit the full potential. For example, [7] calculated the minimum-energy trajectory for a hexarotor maneuver based on models of the vehicle rotors and motors, while [4] and [8] performed similar research for quadrotor configurations. In [9], a quadrotor energy-optimal control problem is defined and solved which considers motor and rigid-body dynamics. In [5], the performance of a quadrotor is evaluated in horizontal flight by calculating the energy consumed in a constant-altitude forward flight operation. Results are compared for minimum acceleration, jerk, and snap operations. Finally, in [10], energy-optimal trajectories are determined using a multirotor model which considers rigid body, motor, and battery dynamics. Notably, propeller aerodynamics are often disregarded in existing works, with rotor thrust and torque commonly approximated as proportional to the square of angular velocity, as in [4], [7], [8], [9], and [10]. This approximation significantly affects the energy performance due to the neglected effect of vehicle motion on rotor behavior, as demonstrated in [11].

To address the gap in the state of the art, this paper builds on our prior modeling and control works presented in [12], [13]. In [12] and [13], a system-level model of the multi-physical dynamics of an octorotor vehicle, including rotor aerodynamics, motor and electronic speed controller (ESC) electro-mechanical dynamics, battery electrical dynamics, and airframe rigid-body dynamics, was formulated, parameterized, validated, and analyzed. The model was then integrated with a simplified quadrotor rigid-body dynamics model in [11], and used for time-optimal and energy-optimal control in a hover-to-hover 100-meter forward-flight operation. In this work, the quadrotor model is further extended to a full octorotor model, which accurately captures the behavior of an experimental vehicle used for testing. Then, a trajectory-optimization methodology is defined based on the model, and used to generate energy-optimal solutions for a range of hover-to-hover operations, which include both vertical and horizontal components. The features of the generated trajectories are described and examined based on key model-based vehicle performance behaviors. Next, a process is described to generate energy-efficient state trajectories for a given operation with minimal computational requirements, which can allow for flexible and efficient real-time trajectory calculation. In this process, the optimized

trajectories are segmented and normalized according to the features observed, and polynomial fits are obtained for each segment. Finally, a feedback control configuration is used in simulation to test the polynomial trajectories against the original optimized trajectories, and a simple baseline feedback controller is also defined for comparison. The energy performance of each control approach is evaluated across a range of operations, and the performance and behavior of each controller for a sample operation are described in detail to illustrate the benefits of model-based control.

II. MODELING

In order to properly capture the complete physical dynamics of the octorotor UAV, a system-level model is used. This model considers the aerodynamics of the propeller-rotor assembly, the electro-mechanical dynamics of the motor and the ESC, electrical dynamics of the battery, and the rigid body dynamics of the vehicle, as well as the integration of each component into the full model. The complete model derivations, as well as the parameterization process and results, can be found in [12]. An overview of certain model behaviors which affect the performance of the vehicle, i.e. efficient trajectory generation in the following sections, is provided in this section.

A. Propeller Aerodynamics

Propeller aerodynamics are modeled using the blade element momentum theory to calculate the forces generated for propulsion [14]. This model is established in two parts, with thrust T and torque Q calculated with the blade element theory, which itself uses the induced air velocity v_i calculated by the momentum theory. The blade element theory model also uses the rotor angular velocity ω , from the motor model in section II-B, and the rotor horizontal and vertical velocities v_x and v_z from the rigid-body dynamics model in section II-C. Using these inputs, the element lift dL and drag dD for each infinitesimal segment of a rotor blade (at distance r from the rotor hub and angle ψ along the revolution of the blade), as shown in Fig. 1, are calculated and converted to element thrust dT and torque dQ of the total N blades as

$$\begin{aligned} dL &= 0.5\rho u^2 c_l dr \\ dD &= 0.5\rho u^2 c_d dr \\ dT &= N(dL \cos \phi - dD \sin \phi) \\ dQ &= Nr(dL \sin \phi + dD \cos \phi). \end{aligned} \quad (1)$$

Here, ρ is the air density, c is the chord length, and c_l and c_d are the lift and drag coefficients. Additionally, the segment inflow velocity u and the inflow angle ϕ (relative to the rotor

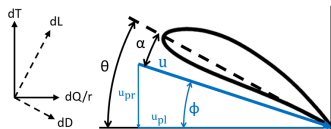


Fig. 1. Geometries of a Blade Cross-section

disk) are calculated based on the planar and perpendicular inflow components u_{pl} and u_{pr} as

$$\begin{aligned} u_{pl}(r, \psi) &= \omega r + v_x \sin(\psi) \\ u_{pr}(r) &= v_i + v_z \\ u(r, \psi) &= \sqrt{u_{pl}^2 + u_{pr}^2} \\ \phi(r, \psi) &= \tan^{-1}(u_{pr}/u_{pl}). \end{aligned} \quad (2)$$

It is also noted that c_d is treated as a constant, while c_l is linearly related to the aerodynamic angle of attack α between ϕ and the section zero-lift line θ . Integration of dT and dQ over r gives the final equations for T and Q as

$$\begin{aligned} T &= \int_{R_0}^{0.97R} \int_0^{2\pi} N \rho u_{pl}^2 c_a (\theta - u_{pr}/u_{pl}) d\psi dr / 4\pi \\ Q &= \int_{R_0}^R \int_0^{2\pi} N r \rho u_{pl}^2 c (\phi a (\theta - u_{pr}/u_{pl}) + c_d) d\psi dr / 4\pi. \end{aligned} \quad (3)$$

The integration is performed from the base of the blade R_0 to 97% of the tip R (instead of 100% to approximate the tip loss) and averaged over one full revolution. These equations are also simplified by the small-angle approximation and the assumptions that $u_{pr} \ll u_{pl}$ and $dD \ll dL$, as validated in [12].

To complete the blade element force calculations, v_i must be found using the momentum theory [14]. This theory models the airflow across the rotor as a steady stream. Stationary air enters the stream with relative velocity components v_x and v_z due to the motion of the rotor itself. The horizontal component is constant throughout the stream, while the vertical component increases to $v_z + v_i$ at the rotor disk and $v_z + v_o$ at the stream outlet. Both v_i and v_o are assumed to be uniform over the rotor disk and at the stream outlet, and by combining conservation of momentum and of kinetic energy, v_o is found to be twice of v_i . Therefore, using conservation of momentum, T can be found in terms of the mass flow rate $\dot{m}$ and rearranged as

$$\begin{aligned} \dot{m} &= \rho \pi R^2 \sqrt{v_x^2 + (v_z + v_i)^2} \\ T &= \dot{m} v_o = 2\dot{m} v_i = 2\rho \pi R^2 v_i \sqrt{v_x^2 + (v_z + v_i)^2} \\ v_i^4 + 2v_z v_i^3 + v_z^2 v_i^2 &= (T/2\rho \pi R^2)^2. \end{aligned} \quad (4)$$

This equation can be solved iteratively to calculate v_i using the value of T found from blade element theory.

B. Motor Assembly Electro-Mechanical Dynamics

To calculate the angular velocity ω and the current draw I_b for each rotor, a model of the electro-mechanical dynamics of the motor-ESC system is needed. The ESC model calculates I_b as well as the motor input voltage V_{in} , which is used in the motor model along with motor/propeller torque Q from section II-A to find ω and the motor input current I_{in} . The inputs to the ESC model are the battery voltage V_b , the PWM (Pulse-Width Modulation) command, and I_{in} .

Specifically, ESC is treated as a transformer, transmitting a portion of V_b based on PWM according to

$$V_{in} = V_b F_{ESC\%}(PWM), \quad (5)$$

where $F_{ESC\%}(PWM)$ is a nonlinear function relating PWM to the percentage of voltage transmitted. The brushless DC motor is then modeled based on the commonly used 3-parameter model [15]–[17], disregarding transient effects due to the rapid response time of the motor used. In this model, the three phases of the motor are treated as one equivalent circuit. First, I_m is calculated as

$$I_m = \frac{Q}{K_T} = QK_V, \quad (6)$$

using the motor torque constant K_T , which is equal to the inverse of the motor velocity constant K_V . The motor voltage V_m can then be found as

$$V_m = V_{in} - I_m R_m, \quad (7)$$

by subtracting the voltage drop across the winding resistance R_m , and used to calculate ω as

$$\omega = V_m K_V. \quad (8)$$

Meanwhile, the battery current I_b is determined based on the power balance of the ESC,

$$\begin{aligned} V_b I_b \eta_{ESC} &= V_{in} I_{in} \\ I_b &= V_{in} (I_m + I_0) / (V_b \eta_{ESC}), \end{aligned} \quad (9)$$

where η_{ESC} is the nonlinear ESC efficiency as a function of V_b and PWM. In this model, other motor losses, such as friction and magnetic losses, are included in the motor zero-load current I_0 , which is the current drawn when the torque load is zero [17].

C. Airframe Rigid Body Dynamics

Finally, the rigid body dynamics of the airframe are modeled to calculate the motion of each rotor using the thrusts and torques from the propeller model. Two-dimensional dynamics are considered in this work, although standard equations for three-dimensional dynamics can be found in existing literature [18].

Using the Newton's second law, the linear and angular accelerations of the multirotor are calculated as

$$\begin{aligned} \ddot{X} &= \Sigma T \sin(\Theta) / m - C_{BD} \dot{X} |\dot{X}| / m \\ \ddot{Z} &= \Sigma T \cos(\Theta) / m - g \\ \ddot{\Theta} &= \tau_{\Theta} / J_y \\ &= L_{arm} ((T_5 + T_1 - T_2 - T_4) \cos(\pi/8) \\ &\quad + (T_7 + T_3 - T_6 - T_8) \sin(\pi/8)) / J_y, \end{aligned} \quad (10)$$

which are then integrated over time to obtain the velocity, position, and orientation. Specifically, X and Z are the horizontal and vertical positions of the vehicle center of mass in the global frame, and Θ is the pitch angle. In addition, ΣT is the sum of thrusts of all rotors computed by the propeller model, $C_{BD} \dot{X} |\dot{X}|$ is the body drag force with C_{BD} as the drag coefficient, J_y is the moment of inertia about the y axis, L_{arm} is the arm length from the center of mass to the rotors, and τ_{Θ} is the total pitch-axis torque on the vehicle, with each rotor numbered according to Fig. 2. Notably, as the body drag is proportional to the velocity squared and

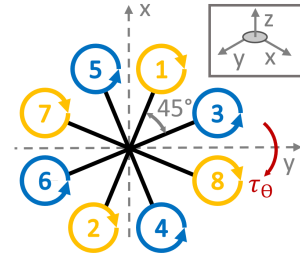


Fig. 2. Layout of Rotors, Axes, and Torques (Inset: Axis Orientations)

the forward velocity tends to be significantly larger than the vertical velocity, vertical drag is not considered. The inflow terms can then be calculated for each rotor as

$$\begin{aligned} v_x &= \dot{X} \cos \Theta + \dot{Z} \sin \Theta \\ v_z &= -\dot{X} \sin \Theta + \dot{Z} \cos \Theta + \dot{\Theta} x_i \end{aligned} \quad (11)$$

where x_i is the x position of the rotor relative to the vehicle center of mass.

D. Model Integration

With each subsystem model defined, the overall system model can be integrated as shown in Fig. 3. The control inputs are the PWM commands for each motor, which instruct the ESC to regulate the motor input voltage as a fraction of the battery voltage. Due to the short duration of the operations considered in this work, the battery electrical dynamics are neglected with the voltage fixed at 25 V, and a fully-integrated battery model will be considered in future work to account for the significant effect of battery behavior on vehicle performance during long flights [12], [11]. The motor responds with a current draw and a rotational speed, which also depends on the torque load from the propeller. The motor rotation drives the propeller to generate the torque and thrust per blade element theory. Thrust and torque are used in the rigid body dynamics model to calculate the motion of the UAV, which affects the propeller velocities for both blade element and momentum theories. Finally, the torque is looped back to the motor and determines the motor speed and current, which is further looped back to ESC to determine the current drawn from the battery. The current draw is treated as an output of the system, and can be multiplied by the battery voltage to calculate the power consumed during the operation. Noted that a battery model can be further incorporated to account for the battery electrical and thermal dynamics and their interaction with other subsystem dynamics [12], [19], [20], and the effect of wind can be included through modeling as well [21].

E. Model-Based Vehicle Performance Behaviors

Based on the model equations, key performance behaviors can be observed that will have a significant impact on the energy-efficient trajectories discussed in the following sections. First, it can be shown that rotor efficiency, defined as thrust generated per unit power input, decreases as ω increases. According to Eqn. 2 and Eqn. 3, both T

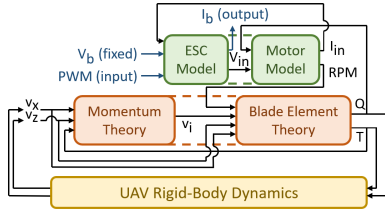


Fig. 3. Block Diagram of Integrated System Model

and Q are approximately proportional to the square of ω [12]. Consequently, the mechanical power output from the motor to the rotor ($Q\omega$) increases with ω more rapidly than thrust. This effect is especially pronounced at high angular velocities, where a small increase in thrust demanded can cause a disproportionately larger increase in power consumption, hence reducing the energy efficiency. Second, the effect of vehicle-frame horizontal velocity v_x and vertical velocity v_z on rotor efficiency can be derived. Increasing v_z generally decreases rotor efficiency, primarily due to the relationship between the aerodynamic angle of attack α and the lift coefficient c_l . As u_{pr} increases, α and c_l become smaller, and the lift-to-drag ratio L/D decreases accordingly, reducing the thrust generation and hence rotor efficiency. Meanwhile, increasing v_x generally has a positive effect on rotor efficiency by increasing the root-mean-squared u_{pl} , which elevates the average lift-to-drag ratio over a full rotor revolution. It should also be noted that such effect will be reduced at high forward velocity. As forward velocity increases, the vehicle must pitch forward significantly to provide acceleration and overcome drag. High pitch angle negatively affects rotor efficiency, as the component of the forward velocity perpendicular to the rotor disk becomes larger, hence increasing v_z .

III. OPTIMIZED TRAJECTORY GENERATION

The developed system-level dynamic model of the vehicle is applied to UAV control in this section through model-based optimization of state trajectories. Optimization results for a range of operations are presented, and key features and behaviors are discussed.

A. Optimization Methodology

To generate the optimized trajectories, energy-optimal control is formulated according to the cost function

$$E_c = \int_0^{t_f} V_b \Sigma I dt, \quad (12)$$

where ΣI is the sum of the current drawn by each ESC (rotor) and t_f is the end time of the operation. The system is defined in the form of state-space equations, with the state vector $\tilde{\mathbf{x}} = [X, \dot{X}, Z, \dot{Z}, \Theta, \dot{\Theta}]^T$. The operation begins with the vehicle in stationary hover at the origin, and ends in stationary hover at the targeted endpoint with forward position X_f and vertical position Z_f ,

$$\tilde{\mathbf{x}}(t_f) = [X_f, 0, Z_f, 0, 0, 0]^T \quad (13)$$

The control inputs u_1 through u_4 are defined as

$$\begin{aligned} u_1 &= \ddot{Z} = \Sigma T \cos(\Theta)/m - g, & u_2 &= \ddot{\Theta} = \tau_{\Theta}/J_y \\ u_3 &= T_{3,7} - \Sigma T/8, & u_4 &= T_{6,8} - \Sigma T/8, \end{aligned} \quad (14)$$

where $T_{3,7}$ and $T_{6,8}$ are the thrusts of the middle-front and middle-rear rotors, respectively. The first two inputs emulate a typical piloted flight control mode, where the accelerations of vertical and pitching motion are set explicitly. The third and fourth inputs are used to balance the thrust efficiently across the rotors. For two-dimensional operations (along x and z), the rotor thrusts are paired to be symmetric across the x -axis, with zero total torque along the vehicle y - and z -axis. Under this simplification, any arbitrary combination of four input values corresponds to a unique set of thrust values for each of the four rotor pairs. This configuration also allows us to directly impose constraints on the inputs, e.g.

$$\begin{aligned} -2m/s^2 &\leq u_1 \leq 2m/s^2, & -9\text{rad}/s^2 &\leq u_2 \leq 9\text{rad}/s^2, \\ -1.5N &\leq u_3 \leq 1.5N, & -1.5N &\leq u_4 \leq 1.5N, \end{aligned} \quad (15)$$

in order to ensure safe and achievable trajectories. The state derivative with respect to time can then be calculated based on Eqns. 10 and 14,

$$\dot{\tilde{\mathbf{x}}} = [\mathbf{x}_2, (u_1 + g) \tan(\mathbf{x}_5) - C_{BD} \mathbf{x}_2 |\mathbf{x}_2|/m, \mathbf{x}_4, u_1, \mathbf{x}_6, u_2]^T. \quad (16)$$

Finally, to calculate the rotor current, the thrust of each rotor pair needs to be found. From Eqn. 14, ΣT is dictated by u_1 and Θ , meaning $T_{3,7}$ and $T_{6,8}$ can be found for any combination of states and control inputs. Then, based on Eqn. 10, exactly one pair of values for $T_{1,5}$ and $T_{2,4}$ exist which correspond to both ΣT and τ_{Θ} set by u_2 . Prior to optimization, tables of T and I_b as dependent on PWM , V_b , v_x , and v_z are first created by resolving the system dynamics,

$$\begin{aligned} T &= T_{table}(PWM, V_b, v_x, v_z) \\ I &= I_{table}(PWM, V_b, v_x, v_z). \end{aligned} \quad (17)$$

By using these tables, optimized solutions and the associated power consumption (based on current) can be found without resolving the loop of dynamics in Fig. 3 repeatedly over the iterative optimization process, significantly reducing the computational complexity.

B. Optimal Trajectories

Using the state-space equations, control inputs, and cost function described above, solutions have been generated for a range of operations in forward and diagonal climbing flight with the GPOPS-II optimal control toolbox for MATLAB [22]. Specifically, as shown in Fig. 4, solutions were found for each combination of forward displacement from 40 to 70 m in increments of 10 m and vertical displacement from 0 to 44 m in increments of 4 m, with some sets of data removed as outliers where the solution was not properly resolved. Results of sample operations are shown in Fig. 5. Several notable behaviors can be observed from the results. First, the operations are clearly divided into two groups due to the vertical acceleration constraint. For group 1, the time

to complete each operation is primarily defined by the final horizontal displacement X_f . For group 2, which includes operations with larger vertical components, t_f is instead primarily defined by the final vertical displacement Z_f . The groups are further illustrated in Fig. 6, which shows the value of t_f for each combination of X_f and Z_f . As shown, for group 1, t_f increases with X_f but decreases with Z_f . For group 2, t_f can be approximated by $\sqrt{2Z_f}$, which is the minimum time needed to reach Z_f subject to the constraint on $\ddot{Z}$ in Eqn. 15. These relationships are discussed in more detail subsequently.

For optimal trajectories in group 1, as shown in Fig. 5, initially the pitch angle rapidly increases to the (negative) maximum allowed value, and holds briefly to provide maximum forward acceleration. The pitch angle then begins to gradually decrease and turn to the other direction for the remaining operation, as the vehicle reaches a peak forward velocity and starts to decelerate. Finally, as the pitch angle reaches the allowed positive maximum, the vehicle brakes sharply to stabilize at the target endpoint. As a result, the forward position and velocity trajectories are very similar at the start and end of each operation within this group. Meanwhile, the peak forward velocity increases with X_f , and the low velocity in shorter flights reduces the energy used to accelerate the vehicle as higher velocity is not needed. The gradual deceleration period is also extended accordingly as X_f increases, both to maintain a high forward velocity to reach the endpoint more quickly, and to allow body drag to supplement active braking. For the vertical state trajectories, the operation typically begins with a brief drop as the angular acceleration in pitch angle is prioritized over thrust. This short period of negative vertical acceleration is followed by a longer period of high vertical acceleration, and then a gradual deceleration to the endpoint. Similar to the horizontal X motion, the peak vertical velocity increases with Z_f , ensuring that the horizontal and vertical motion are coordinated. However, to improve rotor efficiency, a high $\dot{Z}$ is only held for a short duration, limiting its contribution to v_z according to Eqn. 11. After peaking, vertical velocity also fluctuates, as thrust is adjusted according to other states as well as variations in rotor efficiency throughout the operation.

By contrast, in group 2, the vertical acceleration is held at positive maximum for the first half of the operation and negative maximum (deceleration) for the second half, allowing the vehicle to reach Z_f as rapidly as possible. Although the high vertical velocity negatively impacts rotor efficiency, a slower approach would use more energy due to the increase in operation time. Consequently, the pitch angle Θ and angular velocity $\dot{\Theta}$ behave similarly to group 1 at the

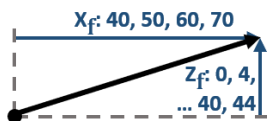


Fig. 4. Diagram of Flight with X and Z Components

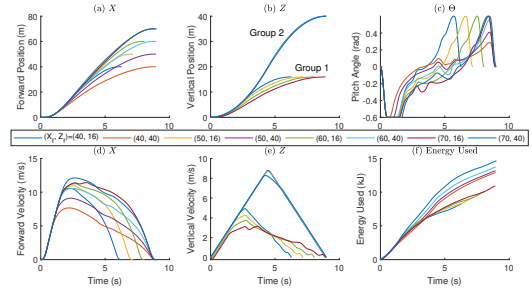


Fig. 5. Trajectories of UAV States and Cumulative Energy Consumption for Sample Optimization Results

beginning, but the time spent in forward acceleration (under maximum negative Θ) decreases with Z_f , to reduce the maximum $\dot{X}$ and coordinate with the vertical Z motion. The forward deceleration period is extended over the increased duration, and the maximum pitch angle as vehicle brakes to reach X_f is reduced. These changes to the forward trajectories, compared to group 1, allow the vehicle to spend less time at extreme pitch angles, reducing the thrust demand of the operation and improving rotor efficiency. Based on the analysis in II-E, reducing the thrust required for forward acceleration is especially valuable for group 2, as the thrust required for the vertical motion of the vehicle is already high, and rotor efficiency is typically low in this case due to the high vertical velocities.

IV. TRAJECTORY FITTING

As the optimal control problem is nonlinear with a large number of variables, the optimization process requires significant time and computational resources, and hence it is not practical for real-time generation of trajectories in real-world operation. To resolve this issue, we propose to use polynomial equations to characterize the optimal state trajectories, which can be used to rapidly generate trajectories for any target combination of X_f and Z_f . To establish the polynomials, the optimized trajectories obtained in the previous section are divided into segments. The segments and their boundaries are defined for trajectories in each group according to the behaviors/features observed, and segments are normalized and averaged as needed to extract the features across trajectories. Finally, after these analyses and pre-processing, polynomials are fit to characterize each segment.

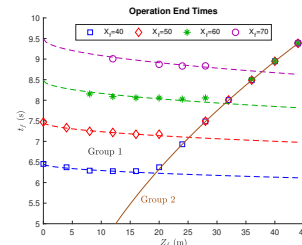


Fig. 6. End Time of Optimized Trajectories Compared to Predictions

Due to page limit constraint, we only present the procedures for group 1 trajectories in detail.

The first step is to predict the operation end time t_f for group 1 and group 2 trajectories as

$$\begin{aligned} t_{f,1} &= (0.102X_f + 2.39)(1 - 0.0198X_f\sqrt{Z_f}) \\ t_{f,2} &= \sqrt{2Z_f}. \end{aligned} \quad (18)$$

The equations are established based on Fig. 6, which shows the relationship between the end time and displacements of the end target point. Specifically, for $t_{f,1}$, the first term of the equation captures the linear dependence of t_f on X_f in group 1, while the second term accounts for the Z_f dependence which has a greater impact in longer operations with larger X_f . The $t_{f,2}$ equation captures the dependence of t_f on $\sqrt{Z_f}$ for the group 2 trajectories as shown in the figure. For a given operation with arbitrary X_f and Z_f , both equations will be applied to calculate t_f , and the larger of the two is adopted with the operation assigned to the according group, which is also illustrated in Fig. 6.

Next, the trajectories of various UAV states are divided into segments according to the behaviors noted in Section III-B. If the operation is part of group 1, the Θ and $\dot{\Theta}$ trajectories are divided into three segments. The first segment is the initial rapid acceleration period, which ends when the pitch angle Θ begins to decrease from the (negative) maximum limit. In the second segment, Θ starts to increase gradually in the positive direction, as $\dot{X}$ peaks and begins to decelerate. In the third and final segment, Θ increases more rapidly, peaks at the upper limit, and finally sharply returns to zero as the vehicle brakes to reach the endpoint. The duration of the first segment increases linearly with X_f , providing more acceleration in longer flights, while the second segment ends at $t_f - 3.2s$, which is observed consistently from all trajectories in the group. Using such segmentation, the Θ and $\dot{\Theta}$ behaviors for the first and third segments are nearly uniform across all operations, and similarly consistent for the second segment after being normalized with respect to time to account for the varying length of this segment among operations of different duration. For X and $\dot{X}$, similar divisions are used, with the notable difference that the $\dot{X}$ trajectory includes another division at the point of maximum forward velocity. This is necessary for accurate normalization, as $\dot{X}$ is non-monotonic during the middle portion of the trajectory. The trajectories for Z and $\dot{Z}$ are relatively complicated, requiring six segments to accurately capture key behaviors. For instance, as Θ peaks near the end of an operation, vertical acceleration decreases temporarily due to the increased energy cost of vertical thrust at extreme pitch angles. To isolate this behavior, the sixth segment for Z and $\dot{Z}$ contains the last 2 seconds of the operation.

An example of the fitted trajectory for one operation ($X_f = 60m, Z_f = 20m$) is shown in Fig. 7, with the segments for Θ indicated and numbered in Fig. 7 (c). The polynomial fits effectively capture the main behavior of the optimized trajectory, although there are some visible discrepancies. In the middle segment of Θ and $\dot{\Theta}$, oscillations are observed. The oscillations are not consistent

across operations, making it difficult to accurately explain and define them, and may be due to numerical issues of the optimizer rather than valid model behaviors. The peak value of $\dot{Z}$ is also difficult to exactly predict, as is some of the behavior in segments 3, 4, and 5 of this trajectory, in part because these segments often coincide with the pitch oscillations. However, the discrepancies are relatively minor, and so are not corrected for in the fitted trajectories to avoid overfitting and to smooth out potential errors in the optimized trajectories. An analysis of simulated results in section V shows that failing to include these does not have a significantly detrimental impact on overall performance.

V. PERFORMANCE EVALUATION OF FITTED TRAJECTORIES

In this section, the generated trajectories are applied to UAV flight control in simulation to validate the energy performance. The results based on the polynomial-fit trajectories are tested and compared to those based on the original optimized trajectories as well as a simple baseline controller. Detailed results for sample operations and averaged results for each group are presented and analyzed.

A. Feedback Control Architecture

In order to implement the generated target trajectories, a PID control structure is used as shown in Fig. 8. Specifically, the control architecture consists of a control script, which includes the target trajectory generator and a first pair of PID controllers, and an onboard controller with a second pair of PID. At each time step, the target values for X , $\dot{X}$, Θ , Z , and $\dot{Z}$, denoted with the subscript t , are provided by the generated target trajectories to the controllers along with the corresponding vehicle states measured as feedback. The trajectories of pitch angle Θ and velocities are needed due to their significant impact on energy performance, and are supplemented with position trajectories to ensure that the vehicle reach and stabilize at the correct end states. The first pair of controllers convert the errors in states to Θ and $\dot{Z}$ commands, denoted with the subscript C , which are fed to the second pair of controllers to regulate $\dot{\Theta}$ and throttle (i.e. total thrust). A set of mixer equations finally translate the commands to the individual motor PWM values, which are the lowest level control inputs applied to the UAV.

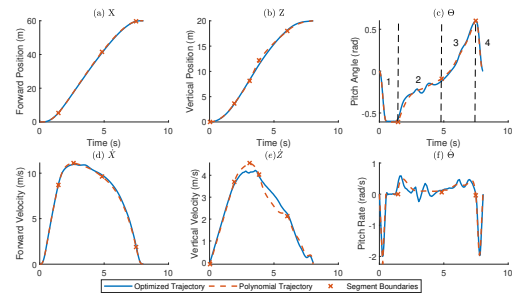


Fig. 7. Comparison of Optimized Trajectories with Polynomial Fits for $X_f = 60, Z_f = 20$, with Segments Indicated in Subplot (c)

This feedback architecture emulates the common manual flight control mode, in which a pilot sends the intermediate commands to the vehicle onboard controller. It should also be noted that roll and yaw control can be easily added to stabilize these states if needed.

In addition, a simple baseline controller is defined for comparison, which does not use a trajectory generator and totally relies on feedback control. A similar PID configuration is adopted, with only $\dot{X}_t$ and $\dot{Z}_t$ used as targets, which are set to be linearly proportional to the remaining distance along each direction. To ensure that the vehicle remains stable and the commands are within the operational range of the vehicle, the following constraints are set,

$$\dot{X}_t < 18m/s, \quad \dot{Z}_t < 5m/s, \quad |\Theta_C| < 0.6rad. \quad (19)$$

B. Results and Discussion

Results are summarized in Table I. Each operation was tested under the original optimized trajectory with feedback control, the polynomial-fit trajectory with feedback control, and the baseline controller. Detailed results are shown for a selected sample of operations, including the improvement/deterioration of energy consumption for each case as compared to that of the original optimized trajectory without feedback (direct optimization results), in the columns “ $\pm\%$ in E”. For each controller, the comparison of average performance for each group and all operations is also provided. From these results, it can be seen that feedback control based on the polynomial trajectories has similar performance to that based on the original optimized trajectories (on average a 0.6% difference in energy consumption), and to the optimized trajectories themselves (on average a 0.7% difference). Meanwhile, the baseline controller consistently has a significantly higher energy cost, on average 11.5% more than the polynomial-fit feedback. From the results of individual operations, it is also observed that the performance of each controller is relatively consistent with slight variation across operations and groups. It should also be noted that, the operation end point of each feedback control case is defined when the distance to the target, vehicle velocity, and pitch angle are within a certain range around (but not exactly at) zero. As a result, the end time for the feedback controllers is often slightly less than that of the original optimized trajectories, giving slightly reduced energy consumption.

To further illustrate the results, the state trajectories of each controller are compared with the original optimized ones for a sample operation ($X_f = 60m, Z_f = 20m$), as shown

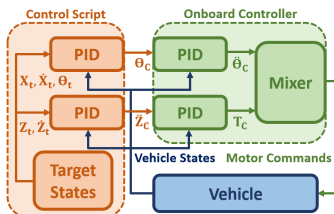


Fig. 8. Feedback Control Architecture

in Fig. 9. It is seen that the trajectory-following feedback controllers remain close to the optimized trajectory, and the slight deviation tends to have only minimal impact on the total energy consumption, as can be observed in subplot (f).

Meanwhile, the significant improvement in performance of the trajectory-following controllers over the baseline controller can be explained by the fact that the former are specifically calibrated based on the vehicle model, hence capable of dynamically balancing the forward and vertical vehicle motion. First, the baseline controller accelerates less efficiently by holding the maximum pitch for an extended period. This results in a much greater energy cost over this time, due to the increased thrust demand and the decrease in rotor efficiency at a steep pitch angle. Moreover, high vertical velocity during this period also further affects the rotor efficiency. By contrast, the optimized trajectory rapidly pitches forward, as the controllers are specifically calibrated based on the vehicle model for a precise maneuver to quickly reach an energy-efficient forward velocity. The pitch angle also begins to decrease before the vertical velocity peaks, reducing energy use during this period. Second, after reaching a high forward velocity during the initial acceleration period, the baseline controller quickly pitches backwards and begins to decelerate sharply. As a result, the vehicle does not remain near the peak velocity long enough to efficiently use the energy spent accelerating, while the optimized trajectory decelerates more gradually to remain near the peak velocity. However, as the vehicle approaches X_f , the baseline controller begins to decelerate more gradually to avoid overshooting, while the optimized trajectory can again adjust its velocity more precisely and efficiently. This allows the optimized trajectory to spend less time, and therefore less energy, in low-velocity flight to reach and stabilize at the endpoint. Finally, the baseline controller rapidly reaches the vertical velocity limit and holds it until nearly reaching Z_f , while the optimized vertical velocity trajectory accelerates and decelerates more gradually. In addition to increasing rotor efficiency and lowering the thrust required, this allows the optimized trajectory to reach X_f and Z_f at the same time, while the baseline reaches Z_f significantly earlier before the operation finishes. In this way, the optimized trajectory uses the model to flexibly balance the vertical and horizontal

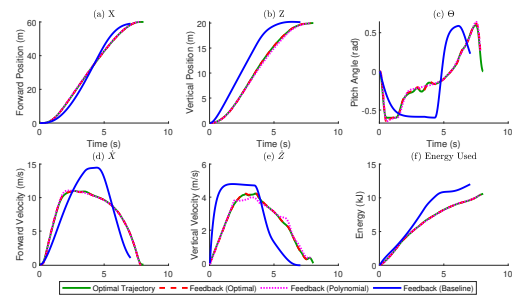


Fig. 9. Comparison of Optimized Trajectories to Feedback Control Results (Using Optimized Trajectories, Polynomial-Fit Trajectories, and Baseline Controller) for $X_f = 60m, Z_f = 20m$

X_f, Z_f (m)	Optimized Profiles		Optimized with Feedback			Polynomial with Feedback			Baseline Feedback		
	Time (s)	Energy (kJ)	t (s)	E (kJ)	$\pm\%$ in E	t (s)	E (kJ)	$\pm\%$ in E	t (s)	E (kJ)	$\pm\%$ in E
50, 0 (Group 1)	7.47	7.00	7.26	6.83	-2.5	7.24	6.88	-1.8	6.88	7.54	+7.7
50, 20 (Group 1)	7.17	9.74	6.97	9.61	-1.3	6.94	9.74	+0.0	7.06	10.97	+12.7
50, 40 (Group 2)	8.95	13.18	8.75	13.07	-0.9	8.78	13.12	-0.5	9.07	14.77	+12.0
70, 0 (Group 1)	9.39	8.45	9.18	8.60	+1.8	9.28	8.57	+1.4	8.15	9.75	+15.5
70, 20 (Group 1)	8.87	11.53	8.67	11.39	-1.2	8.72	11.31	-1.9	8.36	13.48	+16.9
70, 40 (Group 2)	9.01	14.64	8.81	14.54	-0.7	8.80	14.51	-0.9	9.13	16.46	+12.4
Mean (G1, G2)					-1.3% (-1.5%, -1.0%)			-0.7% (-0.7%, -0.5%)			+10.8% (+11.6%, +9.6%)

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT CONTROLLERS

motion, prioritizing each when the associated energy cost is lower, which the baseline controller is unable to achieve based on feedback only. The effects of these behaviors are visible in Fig. 9 (f), which shows how the cumulative energy used by the baseline controller increases much more rapidly near the middle of the flight than that of the optimized trajectories and the two trajectory-following controllers.

VI. CONCLUSIONS

In this paper, a system-level model capturing multi-physical multirotor dynamics is used for energy-efficient control of an octotoror. Based on the model, optimized state trajectories are generated for a range of hover-to-hover flight operations including both forward and climbing components. Based on the behaviors observed, the trajectories were divided into various segments, which are fitted with polynomial approximations. The obtained polynomials can be used to rapidly generate energy-efficient trajectories for any given target endpoint with low computational complexity. The results showed that polynomial approximation performs similarly to the original optimized trajectories, with a mean increase in energy consumption of only 0.6%. Both sets of trajectories showed significant improvement over a baseline controller, which used an average of 11.2% more energy. These results demonstrate the advantages of the proposed model-based control approach, which achieves substantially energy saving through optimization of the trajectories based on the system-level vehicle dynamics.

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